FLOLA

User Guide

v0.1.2

Contents

1. Installation and Launch

1.1 Download

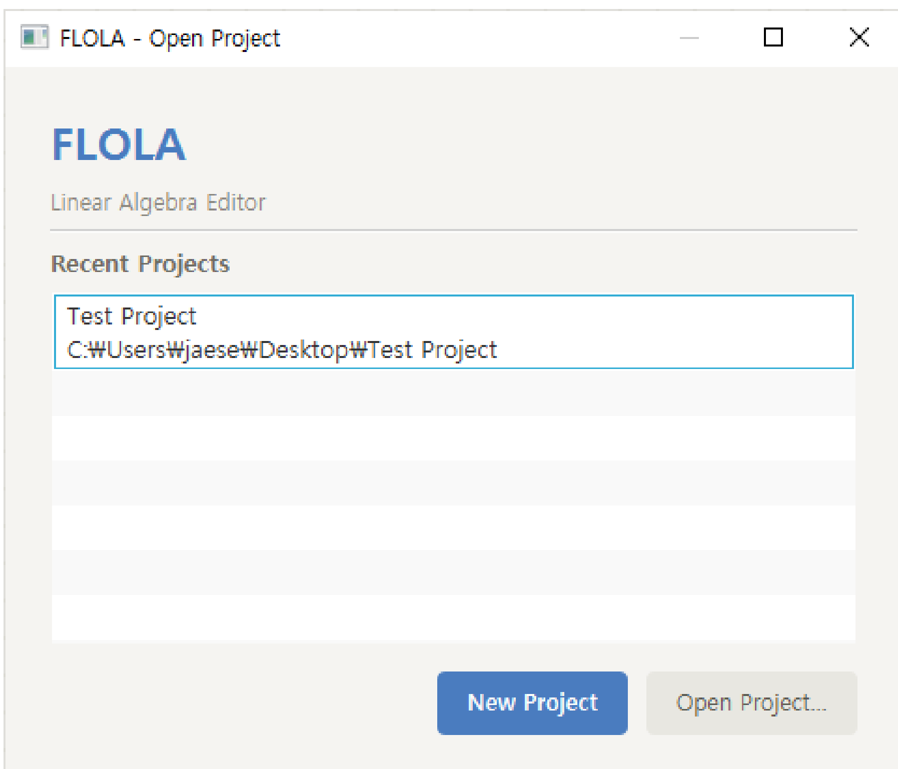
You can download the latest version on the GitHub Releases page.

1.2 Start Screen

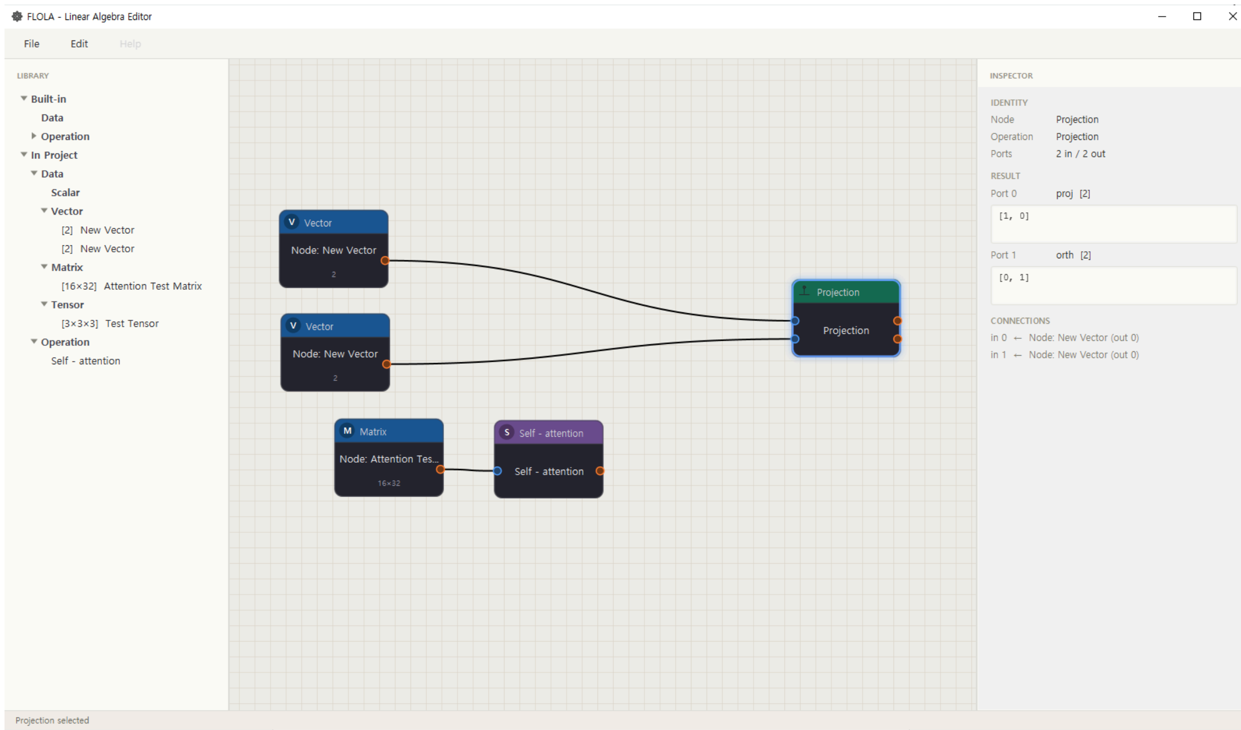
When you launch FLOLA, a project selection window appears first.

- **New Project** — Start with a new, empty project.
- **Open Project…** — Open an existing project by selecting its .flola file directly.
- **Recent Projects** — A list of recently opened projects. Double-click an item to open it.

The recent projects list is saved automatically and shown again the next time you launch the app.



The application is organized into the following areas.



Area	Description
Menu bar	File · Edit · Help menus. Project management and Undo/Redo.
LIBRARY (left)	List of available data and operations. Divided into Built-in and In Project (items created in the current project).
Canvas (center)	The workspace where you place and connect nodes.
INSPECTOR (right)	Shows details of the selected node (name, shape, values, etc.).
Status bar (bottom)	Shows the current status message.

- **Built-in ▸ Data / Operation** — Built-in data and 24 operations. Built-in data presets will be added in a future update.
- **In Project ▸ Data / Operation** — Data and custom operations you created in the current project.

2. Quick Start

A simple example that adds two matrices.

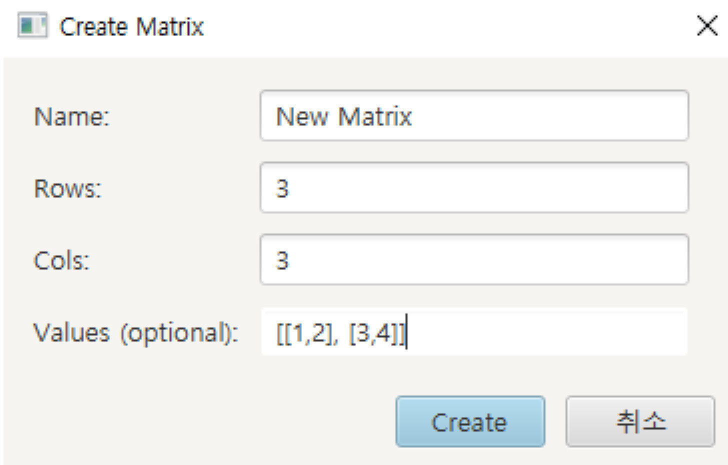
1. In the LIBRARY panel, right-click → New Matrix to create two matrix data items.
2. Double-click a matrix item to open the Tensor Editor. Enter values and save.
3. Drag the matrix item from the LIBRARY onto the canvas.
4. Drag the Add operation from Built-in ▸ Operation onto the canvas.
5. Connect the output port of each matrix node to an input port of the Add node.
6. Double-click the Add node to see the result. (It is also shown in the INSPECTOR panel on the right.)

3. Working with Tensor Data

3.1 Creating Data

Right-click in the LIBRARY panel and choose the type you want from the menu.

Menu	Data created
New Scalar	Scalar (0D, a single value)
New Vector	Vector (1D)
New Matrix	Matrix (2D)
New Tensor	High-dimensional tensor (3D or more)



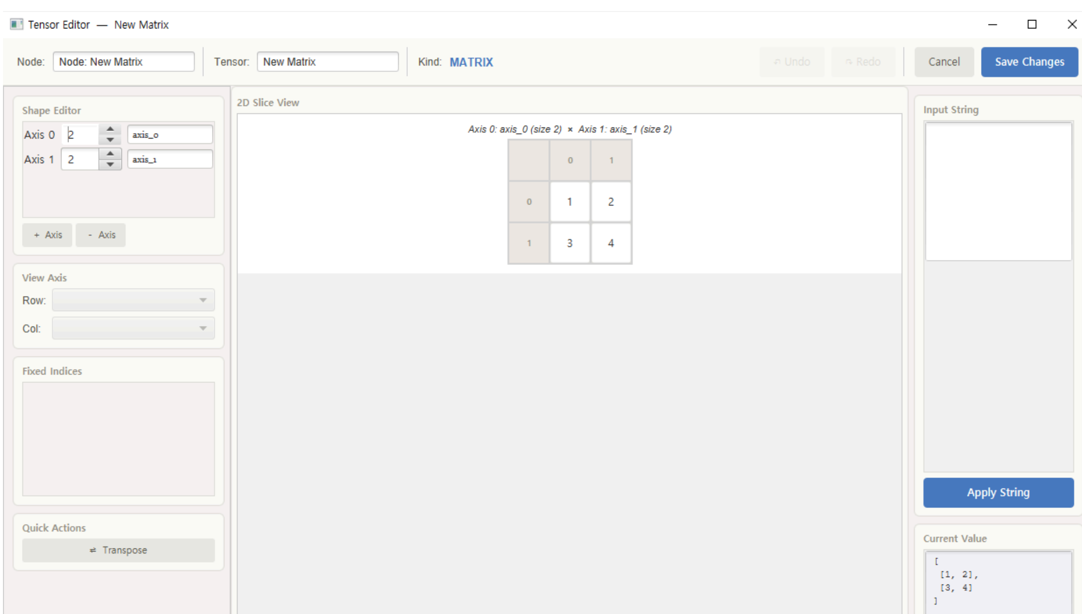
You can create data by entering values in the dialog shown above.

- **Values:** Accepts a list-style string to initialize both the values and the shape at once. In the screenshot above, Row and Col were set to 3, but the data is created from the values and shape of the input string. **When entering, keep the brackets [ ] balanced and the comma separators correct; whitespace is ignored.**

The created data appears under In Project ▸ Data, organized by type.

3.2 Tensor Editor — Editing Values

Double-click a data item or a Tensor node on the canvas to open the Tensor Editor.



- Directly enter and edit each element value
- Change the shape — adjust dimensions and sizes
- Name axes — give each dimension a meaningful name
- Input String — enter a list-style string to edit all values at once
- Undo (Ctrl+Z) / Redo (Ctrl+Y) supported

Changes are saved only when you click Save Changes; any unsaved changes can be discarded with the Cancel button. **(Cancel also resets the Undo/Redo stack)**

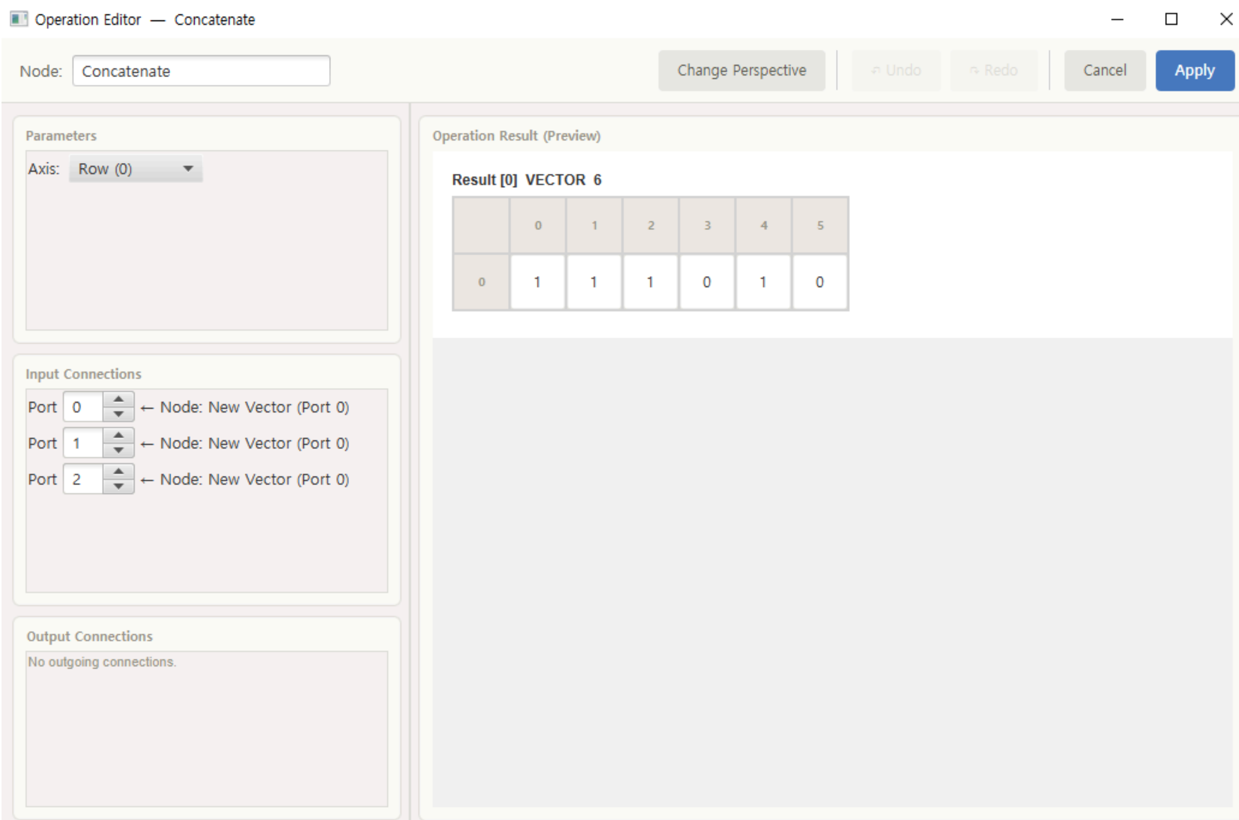
4. Using Operations

4.1 Built-in Operations (24)

Category	Operations
Basic	Add, Subtract, Matrix Multiply, Elementwise Multiply, Divide, Negate, Transpose, Clear, Sum, Average
Activation	ReLU, Sigmoid, Tanh, Softmax
Advanced	Projection, Concatenate, Split, View(reshape), Conv2D, ConvTranspose2D, MaxPool2D, Upsample, SVD, Eigenvalues

4.2 Placing and Configuring Operation Nodes

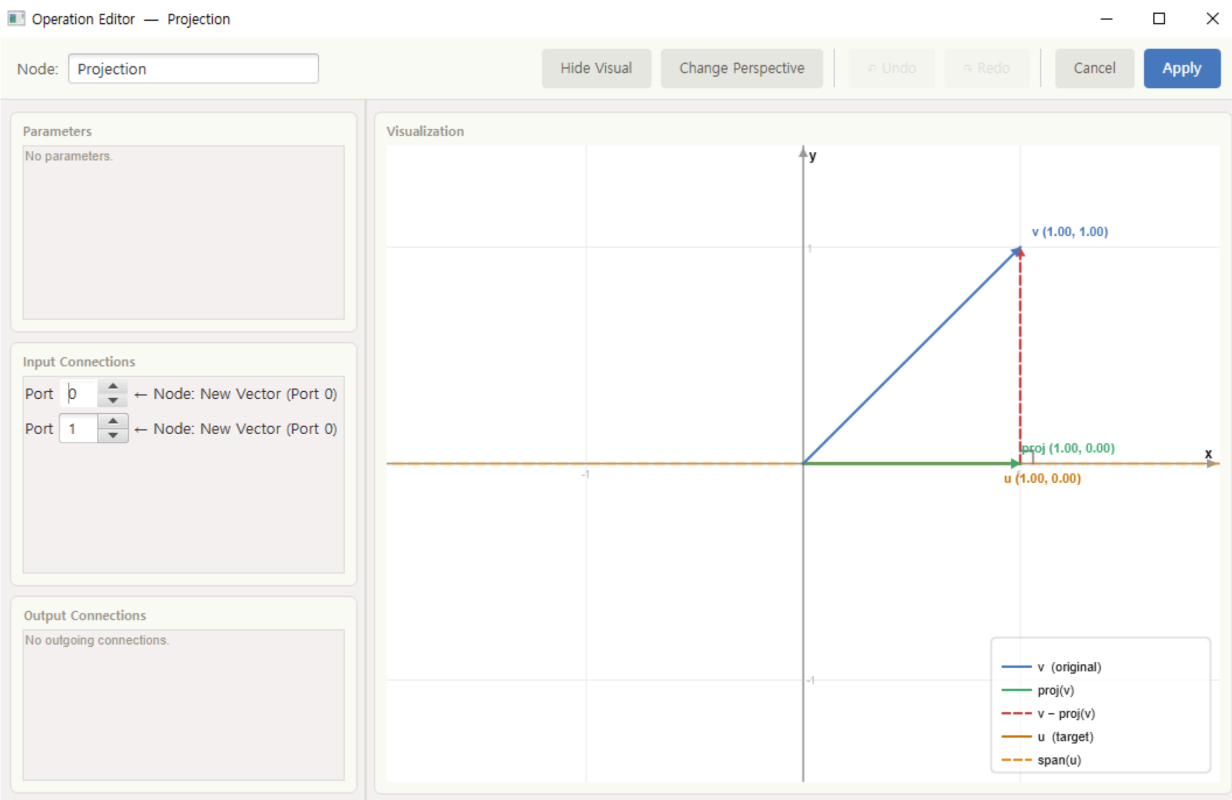
1. Drag the operation you want from Built-in ▸ Operation onto the canvas.
2. Connect the output port of an input node to the input port of the operation node.
3. Some operations need extra settings. Double-click an operation node to open the Operation Editor, where you can adjust parameters (e.g., the kernel size for Conv2D, the target shape for View).
4. The connected nodes and ports are shown and can be changed as needed.
5. Once the connections and settings are complete, the result is computed automatically.



※ If no result appears: no result is produced when an input is missing, the input data is invalid, or the operation cannot be performed. If you think it is a bug, contact me through any of the channels in Contact and I will look into it.

4.3 SVD and Projection Visualization

SVD and Projection provide a visualization feature. Projection is shown only when it can be visualized on a 2-D plane. Click the 'Show Visual' button at the top of the editor to view it.



5. Editing the Graph

5.1 Placing and Connecting Nodes

- **Place** — Drag an item from the LIBRARY onto the canvas.
- **Connect** — Drag one node's **output port** onto another node's **input port**.
- **Disconnect** — Right-Click a connection and delete it, or reconnect a different output to the same input to replace it. (except for combined ports that accept multiple inputs)

5.2 Navigating the Canvas

Action	How to
Zoom in / out	Ctrl + mouse wheel scroll (centered on the cursor)
Pan	Drag with the middle mouse button (wheel)

Action	How to
Select multiple nodes	Drag on empty space to select a region / Ctrl + left-click
Select all	Ctrl + A

5.3 Editing Actions

Action	Shortcut / Method
Move a node	Drag a node (select several to move them together)
Delete	Select node(s), then **Delete** or **Backspace**
Copy / Paste	Ctrl + C / Ctrl + V
Undo / Redo	Ctrl + Z / Ctrl + Shift + Z (or Ctrl + Y)

5.4 Undo / Redo

Almost every edit — adding, deleting, moving, and connecting nodes, editing values, and so on — can be undone and redone across multiple steps.

- Ctrl + Z — Undo
- Ctrl + Shift + Z or Ctrl + Y — Redo

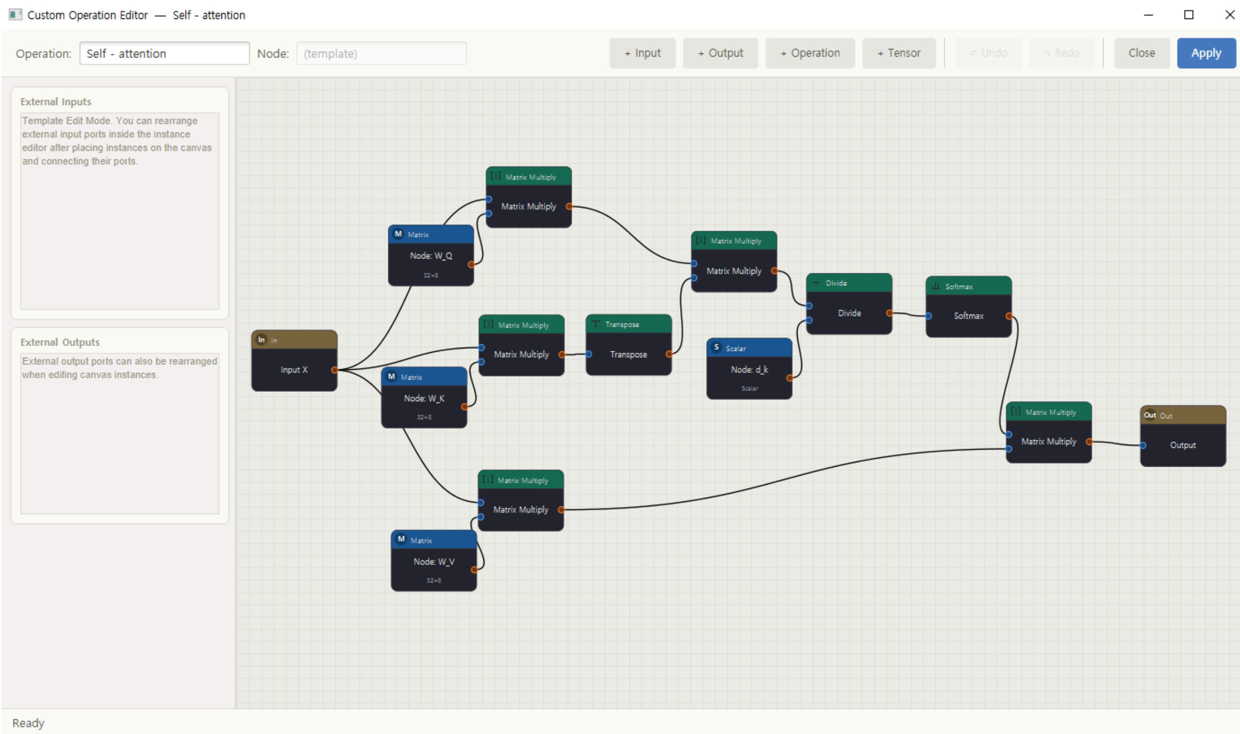
Opening the Edit menu also shows what the next undo/redo will affect (e.g., "Undo: Add Matrix to canvas"), so you can see in advance what will be reverted.

6. Custom Operations

You can save a frequently used set of operations as a single reusable node. Just like the structure of an AI model, complex computations can be encapsulated and reused.

1. In the LIBRARY panel, right-click → New Custom Operation.
2. When the custom operation editor opens, build an operation graph inside it. The points that connect to the outside are defined by **Input nodes and Output nodes**.
3. After you finish and save, it is added as a new custom operation under In Project ▸ Operation.

4. This custom node can be dragged onto the canvas and reused many times, just like a built-in operation. All instances share the same definition, but each one keeps its own computation result.



7. Keyboard Shortcuts

Function	Shortcut
New project	Ctrl + N
Open project	Ctrl + O
Save	Ctrl + S
Undo	Ctrl + Z
Redo	Ctrl + Shift + Z / Ctrl + Y
Copy / Paste	Ctrl + C / Ctrl + V
Select all	Ctrl + A
Delete	Delete

Function	Shortcut
Zoom in / out	Ctrl + mouse wheel
Pan	Drag with the middle mouse button

📬 Contact

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